

## The distributive law



- 1 A student incorrectly used the distributive property and wrote  $6(7x + 2) = 42x + 2$ .

Explain how to correct the error.

- 2 Expand the following expressions:

|                        |                          |                            |                             |
|------------------------|--------------------------|----------------------------|-----------------------------|
| <b>a</b> $5(s + 7)$    | <b>b</b> $9(5 + w)$      | <b>c</b> $3(r - 6)$        | <b>d</b> $(y + 8) \times 7$ |
| <b>e</b> $-8(t + 10)$  | <b>f</b> $-8(c - 5)$     | <b>g</b> $-(y + 2)$        | <b>h</b> $4(4s + 5)$        |
| <b>i</b> $-10(8 + 3r)$ | <b>j</b> $-8(-2t - 3)$   | <b>k</b> $p(q + 4)$        | <b>l</b> $9(u - v)$         |
| <b>m</b> $-(2m - 3n)$  | <b>n</b> $-r(11s + 13t)$ | <b>o</b> $4(2r + 3s + 5t)$ | <b>p</b> $6(4m + 3n + 5p)$  |

- 3 Expand the following expressions:

|                           |                           |                            |                            |
|---------------------------|---------------------------|----------------------------|----------------------------|
| <b>a</b> $6m(2n + 3)$     | <b>b</b> $9(v^2 + 6)$     | <b>c</b> $r(r + 7)$        | <b>d</b> $6m(4m + 7)$      |
| <b>e</b> $-6a(2a + 9)$    | <b>f</b> $4wy(y + w)$     | <b>g</b> $4p(4q - 3 + 6r)$ | <b>h</b> $2f(f - 7g - 6h)$ |
| <b>i</b> $-10(y^2 + 7)$   | <b>j</b> $-(9 - s^3)$     | <b>k</b> $r(s^2 + 8)$      | <b>l</b> $-10(3 + 8u^3)$   |
| <b>m</b> $4s^5(4t^4 - 9)$ | <b>n</b> $6p^3(2q^5 - 9)$ | <b>o</b> $9f(f - 4g - 6h)$ | <b>p</b> $3uv(v + u)$      |

- 4 Expand and simplify the following expressions:

|                              |                            |
|------------------------------|----------------------------|
| <b>a</b> $4(x + 8) - 2$      | <b>b</b> $7 + 5(3x + 4)$   |
| <b>c</b> $5 + 3(x + 4)$      | <b>d</b> $8(3x + 4) - 6x$  |
| <b>e</b> $34 + 9(4x - 5)$    | <b>f</b> $5x - 8(2x - 3)$  |
| <b>g</b> $9x - 5(2x + 3)$    | <b>h</b> $-8(5x - 7) - 9$  |
| <b>i</b> $4y + 5 + 6(y - 9)$ | <b>j</b> $4(5(x - 3) + 2)$ |

- 5 Expand and simplify the following expressions:

|                                      |                                      |
|--------------------------------------|--------------------------------------|
| <b>a</b> $x(x + 4) + 8(x + 9)$       | <b>b</b> $8y^2 - 4y + 7y(y - 5)$     |
| <b>c</b> $3(6x - 5y) - 2(9y + 4x)$   | <b>d</b> $7(y + 9) + 5(y + 6)$       |
| <b>e</b> $10(x - 4) + 3(x + 8)$      | <b>f</b> $5(y + 7) - (y - 9)$        |
| <b>g</b> $-8(x + 4) - 3(x - 2)$      | <b>h</b> $11(a + 4) - 6(4 - a)$      |
| <b>i</b> $4u(6u + 9v) - 2u(8v + 5u)$ | <b>j</b> $9w(8w + 6z) - 5w(4z + 2w)$ |

## Applications

6 A color printer company offers a range of color panels each with a resolution of

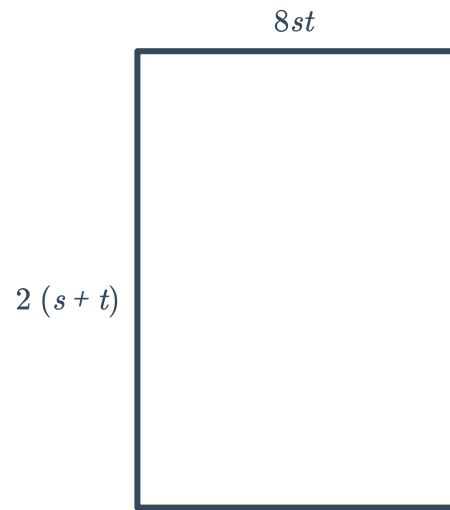
$(m + 8)$  kW, where  $m$  is the panel model number. Write an expression in terms of  $m$  for the total amount of power produced by 11 solar panels.



- 7 The representatives of 8 countries are attending a global summit, and each country is able to send  $(2r - 3)$  representatives. Write an expression for the total number of representatives attending the summit.
- 8 A forklift operator has enough time to make 5 trips between the two ends of a warehouse. A forklift can carry  $(2 + 3n^4)$  boxes at a time. Write an expression for the total number of boxes that can be carried across the warehouse.

- 9 Consider the rectangle shown:

Find an expression for the area of the rectangle in expanded form.



- 10 A rectangle is 8 m longer than it is width. Its width is represented by the pronumeral  $u$ . Form an expression for the area of the rectangle in expanded form.